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Presentation on

RAG-Enhanced Transformer Framework for Context-Aware Polysemy Disambiguation and Automatic Title Generation

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ABSTRACT

AI-Based Polysemy Disambiguation & Automatic Title Generation

1. PROBLEM STATEMENT

Polysemous words can have multiple meanings, making it difficult for NLP systems to correctly understand text context and generate suitable titles.

2. OBJECTIVE

To improve contextual understanding by resolving polysemous words and automatically generating meaningful, context-appropriate titles from textual content.

3. PROPOSED SOLUTION

A RAG-enhanced Transformer framework is proposed to retrieve relevant external knowledge, disambiguate polysemous words, and generate suitable titles based on the given content.

4. EXPECTED OUTCOME

The system is expected to provide better contextual understanding, accurate polysemy disambiguation, and relevant, meaningful titles for diverse textual content.

METHODOLOGY

1. Input Text
2. Preprocessing
3. Identify Ambiguous Words
4. RAG Knowledge Retrieval
5. Transformer Context Analysis
6. Title Generation
7. Evaluation

KEYWORDS | RAG • Transformer • NLP • Polysemy • Contextual Understanding • Disambiguation

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Introduction

01 — Natural Language Processing

NLP enables computers to understand and interpret human language, where meaning depends strongly on context.

02 — Polysemy & Contextual Ambiguity

Polysemous words have multiple meanings, making correct interpretation difficult when context is limited or ambiguous.

03 — Transformer-Based Understanding

Transformers capture relationships between words and provide strong contextual understanding.

04 — RAG-Enhanced Context

RAG retrieves relevant external knowledge to provide additional context for accurate word disambiguation.

05 — Automatic Title Generation

The proposed system combines RAG and Transformers to understand content, resolve ambiguity, and generate meaningful titles.

LITERATURE SURVEY

S.No.	Author(s) & Year	Research / Method Used	Key Findings	Limitations / Research Gap	Proposed Enhancement
1	Bevilacqua et. al. (2021)	Word Sense Disambiguation (WSD) Survey	Reviewed recent advances and representation-learning improvements.	WSD still depends on available sense inventories and contextual resources.	Use external retrieved knowledge to provide richer context.
2	Li et al. (2024)	Transformer + BiLSTM for WSD	Combined Transformer with BiLSTM for Chinee WSb	Focused mainly on Chinese and specific datasets.	Extend disambiguation across diverse domains.
3	Abeysiriwardana & Sumanathilaka (2024)	Lexical Ambiguity Detection and WSD	Reviewed modern approaches for resolving ambiguity.	Scarcity of sense-annotated data and complex texts.	Integrate RAG and broader knowledge sources.
4	Yae et al. (2024/2025)	Large Language Models for WSD	Evaluated LLMs for word sense identification.	LLM performance varies and struggles with ambiguities.	Combine retrieval with LLM reasoning.
5	Zhang et al. (2025)	RoDEval – LLM-based WSD Evaluation	Evaluated LLMs' robustness and disambiguation	LLMs may lack sense coverage and show bias.	Enhance WSD with reliable external knowledge.

Software Requirements	Hardware Requirements
Operating System: Windows	**Processor:** Intel Core i5
Programming Language: Python	**RAM:** Minimum 8 GB
Development Environment: VS Code	**Storage:** Minimum 20 GB available space
NLP & Transformer: Hugging Face Transformers	**GPU:** NVIDIA GPU with at least 4 GB VRAM
RAG Framework: Lang Chain / LlamaIndex	**Internet Connection:** Required for external knowledge retrieval and development resources
Libraries: PyTorch, Pandas, NumPy	**Input/Output Devices:** Keyboard, Mouse and Standard Monitor